

Proof and Induction WS 1

1. Use a direct proof to prove each of the following statements.
 - a. The sum of any two odd integers is even.
 - b. The sum of an odd integer and an even integer is always odd.
2. Use a contrapositive proof to prove each of the following statement: Let n be an integer. If $3n + 2$ is even, then n is even.
3. Use a proof by contradiction to prove each of the following statement: $\sqrt{3}$ is irrational.
4. Prove each of the following logical equivalence: Let n be a positive integer. $n + 9$ is even if and only if $n + 6$ is odd.
5. Consider the following statement: 'If two integers have an even product, then at least one of the two integers must be even.' To prove this statement by contraposition, it would be necessary to:
 - (A) Suppose that at least one of the two integers is even, and then show that the product must be even.
 - (B) Suppose that at least one of the integers is odd, and then show that the product must be odd.
 - (C) Suppose that both integers are odd, and then show that the product must be odd.
 - (D) Suppose that the two integers have an even product and that both integers are odd, and then show that a contradiction arises.
6. Prove that if a, b are integers, then $a^2 - 4b - 3 \neq 0$.
7. For each statement, find the converse and determine if the statement is an equivalence. If it is an equivalence, then write an if statement.
 - a. If a quadrilateral has equal diagonals, then it is a square.
 - b. If $x > 1$, then $\frac{1}{x} < 1$.
 - c. If I pass my exams, then I study hard.
 - d. If $a = 3$, then $a^2 = 9$.
 - e. If a triangle has 2 equal angles, then it is isosceles.
8. Write each mathematical statement in words.
 - a. $\forall m \in \mathbb{N}, \exists n \in \mathbb{Z}: n + m = 0$
 - b. $\forall a, b \in \mathbb{Z}, b \neq 0, \exists p, q \in \mathbb{Q}$ such that $\frac{1}{a+b\sqrt{2}} = p + q\sqrt{2}$
9. Determine the negation of each of the following statement. Also state whether the original statement or the negation is true, justifying your answer where appropriate.
 - a. $\forall x \in \mathbb{R}, x^2 > 0$
 - b. $\exists x \in \mathbb{R}, x^2 = x$
 - c. $\forall n \in \mathbb{Z}^+, 10n > n$
 - d. $\forall x \in \mathbb{R}, x > 0$ or $x < 0$
 - e. $\exists n \in \mathbb{Z}, n \neq 0, n^2 < 1$
 - f. $\forall n \in \mathbb{Z}, (-1)^n = 1$ or $(-1)^n = -1$
10. Mr Eel and Mr Nolah are discussing Mersenne primes which are of the form $2^n - 1$ for some positive integer n .

Mr Eel says, "if n is prime, then $2^n - 1$ must be prime."

Mr Nolah is sure Mr Eel is wrong but is unable to justify his position.

Help Mr Nolah by writing a proof showing Mr Eel is wrong.
11. Prove the following statement by contradiction, or otherwise, for $a, b, c \in \mathbb{Z}$: If $a^2 + b^2 = c^2$, then at least one of a or b is even. You may assume that the square of an odd number is odd, and the square of an even number is even.

12. Consider this statement two positive integers a and b : "If $a^2 + b^2$ is even then either both a and b are even or both a and b are odd".
- Write down the contrapositive of the statement.
 - Prove the original statement by contraposition.
13. If $u_1 = 2, u_2 = 22$ and $u_n = 6u_{n-1} - 5u_{n-2}$ for $n \geq 3$, show that $u_n = 5^n - 3$ for $n \geq 1$.
14. Show that for $n \geq 1, 1 + \frac{x}{1!} + \frac{x(x+1)}{2} + \dots + \frac{x(x+1)\dots(x+n-1)}{n!} = \frac{(x+1)(x+2)\dots(x+n)}{n!}$
15. Prove by induction that $\sin x + \sin 2x + \sin 3x + \dots + \sin nx = \frac{\sin(\frac{nx}{2}) \sin(\frac{(n+1)x}{2})}{\sin(\frac{x}{2})}$ for all counting numbers n . (Hint: You may assume that $2 \cos A \sin B = \sin(A+B) - \sin(A-B)$)
16. If $u_1 = 5$ and $u_n = 3u_{n-1} + 2$ for $n \geq 2$, show that $u_n = 2 \times 3^n - 1$ for $n \geq 1$.
17. Show by Mathematical Induction that $(1 - \alpha_1)(1 - \alpha_2) \dots (1 - \alpha_n) > 1 - (\alpha_1 + \alpha_2 + \dots + \alpha_n)$ for all positive integers n where $n > 1$, if α_k satisfies $0 < \alpha_k < 1$ for $1 \leq k \leq n$.
18. The equation $x^2 - x + 1 = 0$ has roots α and β , and $A_n = \alpha^n + \beta^n$ for $n \geq 1$.
- Without solving the equation, show that $A_1 = 1, A_2 = -1$ and $A_n = A_{n-1} - A_{n-2}$ for $n \geq 3$
 - Hence use induction to show that $A_n = 2 \cos \frac{n\pi}{3}$ for $n \geq 1$
19. The sequence, $1, \sqrt{3}, \sqrt{1+2\sqrt{3}}, \dots$ has its n th position given by $x_n = \sqrt{1+2x_{n-1}}$. By the process of Mathematical Induction, prove that $x_n < 4$ for all $n \geq 1$.
20. In the Fibonacci sequence $1, 1, 2, 3, 5, 8 \dots$ the terms of the sequence are defined recursively by the equation $F_{n+1} = F_n + F_{n-1}$ where $F_1 = 1, F_2 = 1$ and $n \geq 2$. Use induction to prove that, for $n \geq 2$, $F_n^2 - F_{n+1}F_{n-1} = (-1)^{n+1}$.
21. Given the polynomial function: $f_n(x) = 1 + \frac{x}{1!} + \frac{x(x+1)}{2!} + \dots + \frac{x(x+1)\dots(x+n-1)}{n!}$ For $n = 1, 2, 3$.
Where for $n = 1, f_1(x) = 1 + \frac{x}{1!} = x + 1$ which has a zero at -1
- Show that for $n = 2, f_2(x) = \frac{1}{2!}(x+1)(x+2)$ and state the zeroes of $f_2(x)$
 - Hence complete the proof by mathematical induction that the zeroes of the polynomial function $f_n(x)$ are $-1, -2, -3 \dots, -n$ for $n = 1, 2, 3 \dots$, that is, prove that
- $$f_n(x) = \frac{1}{n!}(x+1)(x+2) \dots (x+n)$$
22. Use the principle of mathematical induction to prove $3^n > 2n + 4$, for $n = 2, 3, 4, \dots$
23. Use mathematical induction to prove that $7^n > 4^n + 5^n$ for all integers $n \geq 2$
24. Use mathematical induction to prove that $\frac{1}{2} + \frac{2}{3} + \dots + \frac{n}{n+1} < \frac{n^2}{n+1}$ for all integers $n \geq 2$.
25. Use Mathematical Induction to show that $n! > 2^{n-1}$ for $n = 1, 2, 3, \dots$
26. Use Mathematical Induction to show that $5^n > 3^n + 4^n$ for all positive integers $n \geq 3$.

Answer

5. C
7. a. The quadrilateral is a square then it has equal diagonals. Not an equivalence. Original statement is false; b. If $\frac{1}{x} < 1$ then $x > 1$. Not an equivalence. If $x < 0$ then $\frac{1}{x} < 1$ but $x \not> 1$; c. If I study hard then I pass my exams. Not necessarily true; d. If $a^2 = 9$ then $a = 3$. Not necessarily true; $a = -3$; e. If a triangle is isosceles then it has 2 equal angles. Equivalence. A triangle is isosceles $\Leftrightarrow$ it has two equal angles.
8. a. For all natural numbers m , there exists an integer n such that $n + m = 0$; b. For all integers a and b where b is non-zero, there exist rational number p and q such that $\frac{1}{a+b\sqrt{2}} = p + q\sqrt{2}$.
13. If a and b are two positive integers and one of a and b is even and the other is odd then $a^2 + b^2$ is odd